

9.18 TEST OF SIGNIFICANCE FOR SINGLE PROPORTION - LARGE SAMPLES

[JNTU (H) Nov. 2009 (Set - 4)]

Suppose a large random sample of size n has a sample proportion p of members possessing a certain attribute (i.e., proportion of successes). To test the hypothesis that the proportion P in the population has a specified value P_0 .

Let us set the Null Hypothesis be $H_0 : P = P_0$ (P_0 is a particular value of P)

Then the Alternative Hypothesis is

(i) $H_1 : P \neq P_0$ (i.e., $P > P_0$ or $P < P_0$)

(ii) $H_1 : P > P_0$ or

(iii) $H_1 : P < P_0$

Since n is large, the sampling distribution of p is approximately normal.

$\therefore$ If H_0 is true, the test statistic

$$z = \frac{p - P_0}{\text{S.E. of } p} \text{ or}$$

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}, \text{ where } p \text{ is the sample proportion}$$

is approximately normally distributed.

The critical region for z depending on the nature of H_1 and level of significance α is given in the following table :

Table 9.1 : Rejection Rule for $H_0 : P = P_0$			
Level of significance	1%	5%	10%
Critical region for $P \neq P_0$ (Two-tailed test)	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $P > P_0$ (Right-tailed test)	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $P < P_0$ (Left-tailed test)	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note 1 : Without any reference to the level of significance, we may reject the Null Hypothesis H_0 when $|z| > 3$.

Note 2 : (i) Limits for population proportion P are given by $p \pm 3\sqrt{\frac{pq}{n}}$ where $q = 1 - p$.

(ii) Confidence interval for proportion P for large sample at α level of significance is

$$P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} < P < P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \text{ where } Q = 1 - P \text{ and}$$

$$z_{\alpha/2} = 1.96 \text{ (for 95\%), } z_{\alpha/2} = 2.33 \text{ (for 98\%) and}$$

$$z_{\alpha/2} = 2.58 \text{ (for 99\%).}$$

SOLVED EXAMPLES

Example 1 : A manufacturer claimed that atleast 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test his claim at 5% level of significance.

[JNTU (K), (H) Nov. 2009, (H) Dec. 2011 (Set No. 1)]

Solution : Given sample size, $n = 200$

Number of pieces confirming to specification = $200 - 18 = 182$

$\therefore p$ = Proportion of pieces confirming to specifications

$$= \frac{182}{200} = 0.91$$

and P = Population proportion = $\frac{95}{100} = 0.95$

1. **Null Hypothesis H_0 :** The proportion of pieces confirming to specifications
i.e. $P = 95\%$.

2. **Alternative Hypothesis H_1 :** $P < 0.95$ (left - tail test)

$$3. \text{ The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.91 - 0.95}{\sqrt{\frac{0.95 \times 0.05}{200}}} = \frac{-0.04}{0.0154} = -2.59.$$

Since alternative hypothesis is left tailed, the tabulated value of Z at 5% level of significance is 1.645.

Since calculated value of $|z| = 2.6$ is greater than 1.645, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the manufacture's claim is rejected.

Example 2 : In a sample of 1000 people in Karnataka 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at 1% level of significance ?
[JNTU (K), (H) Nov. 2009, (A) Nov. 2010 (Set No. 3)]

Solution :

Given $n = 1000$

$$p = \text{Sample proportion of rice eaters} = \frac{540}{1000} = 0.54$$

$$P = \text{Population proportion of rice eaters} = \frac{1}{2} = 0.5$$

$$\therefore Q = 0.5$$

Null Hypothesis H_0 : Both rice and wheat are equally popular in the state.

Alternative Hypothesis H_1 : $P \neq 0.5$ (two - tailed alternative)

$$\text{Test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = 2.532$$

The calculated value of $z = 2.532$

The tabulated value of z at 1% level of significance for two-tailed test is 2.58.

Since calculated $z <$ tabulated z , we accept the Null Hypothesis H_0 at 1% level of significance and conclude that both rice and wheat are equally popular in the state.

Example 3 : In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers ?

[JNTU (K) Nov. 2009, (A) Nov. 2010, (K) May 2013 (Set No. 1)]

Solution : Given $n = 600$

Number of smokers = 325

$$p = \text{Sample proportion of smokers} = \frac{325}{600} = 0.5417$$

$$P = \text{Population proportion of smokers in the city} = \frac{1}{2} = 0.5$$

$$Q = 1 - P = 1 - 0.5 = 0.5$$

1. **Null Hypothesis H_0 :** The number of smokers and non-smokers are equal in the city.

2. **Alternative Hypothesis :** $P > 0.5$ (Right tailed)

$$3. \text{ The Test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.5417 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 2.04$$

$\therefore$ Calculated value of $z = 2.04$

Tabulated value of z at 5% level of significance for right tail test is 1.645.

Since, calculated value of $z >$ tabulated value of z , we reject the Null Hypothesis and conclude that the majority of men in the city are smokers.

Example 4 : A die was thrown 9000 times and of these 3220 yielded a 3 or 4. Is this consistent with the hypothesis that the die was unbiased? [JNTU (K) Nov. 2009 (Set No. 3)]

Solution : Given $n = 9000$

$$p = \text{Proportion of successes of getting 3 or 4 in 9000 throws} = \frac{3220}{9000} = 0.3578$$

$P =$ Population proportion of successes

$$= P(\text{getting a 3 or 4}) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} = 0.3333$$

$$\therefore Q = 1 - P = 1 - \frac{1}{3} = \frac{2}{3} = 0.6667$$

1. Null Hypothesis H_0 : The die is unbiased.

2. Alternative Hypothesis H_1 : $P \neq \frac{1}{3}$ (two-tailed)

$$3. \text{ The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.3578 - 0.3333}{\sqrt{\frac{0.3333 \times 0.6667}{9000}}} = 4.94$$

Calculated $z = 4.94$

As $z > 2.58$, the null hypothesis H_0 has to be rejected at 1% level of significance and we conclude that the die is biased.

Example 5 : A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Find the percentage of bad pineapples in the consignment.

Solution :

Given $n = 500$

$$p = \text{Proportion of bad pineapples in the sample} = \frac{65}{500} = 0.13$$

$$q = 1 - p = 0.87$$

We know that the limits for population proportion P are given by

$$p \pm 3 \sqrt{\frac{pq}{n}} = 0.13 \pm 3 \sqrt{\frac{0.13 \times 0.87}{500}} = 0.13 \pm 0.045 = (0.085, 0.175)$$

$\therefore$ The percentage of bad pineapples in the consignment lies between 8.5 and 17.5.

Example 6 : In a random sample of 125 cool drinkers, 68 said they prefer thumsup to pepsi. Test the null hypothesis $P = 0.5$ against the alternative hypothesis $P > 0.5$. [JNTU 2006S (Set No. 2)]

Solution : We have $n = 125$, $x = 68$ and $p = \frac{x}{n} = \frac{68}{125} = 0.544$

1. Null Hypothesis H_0 : $P = 0.5$

2. Alternative Hypothesis H_1 : $P > 0.5$

3. Level of significance : $\alpha = 0.05$ (assumed)

4. The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.544 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{125}}}$

$$= \frac{0.044}{0.045} = 0.9839$$

Since calculated value of $|Z|$ is less than 1.645, we accept the Null Hypothesis H_0 at 5% level of significance.

Example 7 : Experience had shown that 20% of a manufactured product is of the top quality. In one day's production of 400 articles only 50 are of top quality. Test the hypothesis at 0.05 level. [JNTU 2005 S, 2008 S (Set No. 2), (K) May 2013 (Set No. 4)]

Solution : We have $n = 400$, $x = 50$ and $p = \frac{x}{n} = \frac{50}{400} = 0.125$

1. Null Hypothesis $H_0 : P = 0.2$
2. Alternative Hypothesis $H_1 : P \neq 0.2$
3. Level of significance : $\alpha = 0.05$

4. The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

i.e., $z = \frac{0.125 - 0.2}{\sqrt{\frac{(0.2)(0.8)}{400}}} = \frac{-0.075}{0.02} = -3.75$

Since $|z| = 3.75 > 1.96$, we reject the Null Hypothesis H_0 at 5% level of significance.

i.e., $P = 20\%$ is not correct.

Example 8 : A social worker believes that fewer than 25% of the couples in a certain area have ever used any form of birth control. A random sample of 120 couples was contacted. Twenty of them said that they have used. Test the belief of the social worker at 0.05 level.

[JNTU 2005 (Set No. 1)]

Solution : We have

$$n = 120, x = 20, p = \frac{x}{n} = \frac{20}{120} = \frac{1}{6} \text{ and } P = 0.25, Q = 1 - P = 0.75$$

1. Null Hypothesis $H_0 : P = 0.25$
2. Alternative Hypothesis $H_1 : P < 0.25$ (left tailed test)
3. Level of significance, $\alpha = 0.05$
4. The test statistic is

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{\frac{1}{6} - 0.25}{\sqrt{\frac{(0.25)(0.75)}{120}}} = \frac{-0.0833}{0.0395} = -2.107$$

Since $|z| = 2.107 < 2.33 = z_{0.05}$, we accept the null hypothesis H_0 . That is, the claim or belief of social worker is true.

Example 9 : A manufacturer claims that only 4% of his products are defective. A random sample of 500 were taken among which 100 were defective. Test the hypothesis at 0.05 level. [JNTU 2005 (Set No. 2)]

Solution : We have

$$x = 100, n = 500, p = \frac{x}{n} = 0.2 \text{ and } P = 0.04, Q = 1 - P = 0.96$$

1. Null Hypothesis $H_0 : P = 0.04$
2. Alternative Hypothesis $H_1 : P > 0.04$ (Right tailed test)
3. Level of significance : $\alpha = 0.05$
4. The test statistic is

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.2 - 0.04}{\sqrt{\frac{(0.04)(0.96)}{500}}} = \frac{-0.16}{0.00876} = -18.26$$

Since $|z| = 18.26 > 1.645 = z_{0.05}$, we reject the null hypothesis H_0 .

Example 10 : In a sample of 500 from a village in Rajasthan, 280 are found to be wheat eaters and the rest rice eaters. Can we assume that the both articles are equally popular. [JNTU (K) 2009, May 2012 (Set No. 4)]

Solution : Given $n = 500$

$$p = \text{Sample proportion of wheat eaters} = \frac{280}{500} = 0.56$$

$$P = \text{Population proportion of wheat eaters} = \frac{1}{2} = 0.5$$

$$\therefore Q = 1 - P = 1 - 0.5 = 0.5.$$

1. Null Hypothesis H_0 : Both rice and wheat are equally popular in Rajasthan.
2. Alternative Hypothesis $H_1 : P \neq 0.5$
3. Level of significance : $\alpha = 0.01$

$$4. \text{ The test statistic is } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.56 - 0.5}{\sqrt{(0.5)(0.5)/500}} = \frac{0.06}{0.022} = 2.68$$

$\therefore$ The calculated value of $z = 2.68$

The tabulated value of z at 1% level of significance for two-tailed test is 2.58.

Since calculated value of $z >$ tabulated z , we reject the null hypothesis H_0 . i.e., Both rice and wheat are not equally popular in Rajasthan at 1% level of significance.

Example 11 : Among 900 people in a state 90 are found to be chapati eaters. Construct 99% confidence interval for the true proportion. [JNTU 2005 S, (K) May 2010S (Set No. 2)]

Solution : Given $x = 90$, $n = 900$

$$\therefore P = \frac{x}{n} = \frac{90}{900} = \frac{1}{10} = 0.1 \text{ and } Q = 1 - P = 0.9$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{(0.1)(0.9)}{900}} = 0.01$$

Confidence interval is

$$\left(P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}}, P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \right) \text{ or } \left(P - 3 \sqrt{\frac{PQ}{n}}, P + 3 \sqrt{\frac{PQ}{n}} \right)$$

$$\text{i.e., } (0.1 - 0.03, 0.1 + 0.03)$$

$$\text{i.e., } (0.07, 0.13).$$

Example 12 : In a random sample of 160 workers exposed to a certain amount of radiation, 24 experienced some ill effects. Construct a 99% confidence interval for the corresponding true percentage. [JNTU 2004, 2006S (Set No. 1)]

Solution : We have $x = 24$, $n = 160$ and $P = \frac{24}{160} = 0.15$, $Q = 1 - P = 0.85$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.15 \times 0.85}{160}} = 0.028$$

Confidence interval at 99% level of significance is

$$\left(P - 3 \sqrt{\frac{PQ}{n}}, P + 3 \sqrt{\frac{PQ}{n}} \right)$$

$$\text{i.e., } (0.15 - 3 \times 0.028, 0.15 + 3 \times 0.028)$$

$$\text{i.e., } (0.065, 0.234).$$

Example 13 : In a random sample of 400 industrial accidents, it was found that 231 were due atleast partially to unsafe working conditions. Construct a 99% confidence interval for the corresponding true proportion. [JNTU 2006S, (K) May 2010S (Set No. 2)]

Solution : We have $x = 231$, $n = 400$ and $P = \frac{x}{n} = \frac{231}{400} = 0.5775$

$$\therefore Q = 1 - P = 1 - 0.577 = 0.4225$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.5775 \times 0.4225}{400}} = 0.0247$$

Confidence interval is

$$\left(P - 3 \cdot \sqrt{\frac{PQ}{n}}, P + 3 \cdot \sqrt{\frac{PQ}{n}} \right)$$

$$\text{i.e., } (0.5775 - 3 \times 0.0247, 0.5775 + 3 \times 0.0247)$$

$$\text{i.e., } (0.5034, 0.6516).$$

Example 14 : 20 people were attacked by a disease and only 18 survived. Will you reject the hypothesis that the survival rate if attacked by this disease is 85% in favour of the hypothesis that is more at 5% level. [JNTU 2008S, (K) May 2010S, (H) April 2012 (Set No. 1)]

Solution :

$$n = \text{Sample size} = 20$$

$$X = \text{Number of survived people} = 18$$

$$p = \text{Proportion of survived people} = \frac{X}{n} = \frac{18}{20} = 0.9$$

$$P = 0.85$$

$$\therefore Q = 1 - P = 1 - 0.85 = 0.15$$

1. Null Hypothesis $H_0 : P = 0.85$
2. Alternative Hypothesis $H_1 : P > 0.85$ (Right tailed test)
3. Level of significance : $\alpha = 0.5$
4. The test statistic is

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.9 - 0.85}{\sqrt{\frac{0.85 \times 0.15}{20}}} = \frac{0.05}{0.08} = 0.625$$

$$\therefore \text{Calculated } z = 0.625$$

Tabulated z at 5% level of significance, $z_\alpha = 1.645$

Since calculated $z <$ tabulated z , we accept the Null Hypothesis H_0 .

i.e., The proportion of the survived people is 0.85.

Example 15 : If 80 patients are treated with an antibiotic 59 got cured. Find a 99% confidence limits to the true population of cure. [JNTU 2004 S (Set No. 2)]

Solution : $n = 80, x = 59$ and $P = \frac{x}{n} = \frac{59}{80} = 0.7375$

$$\therefore Q = 1 - P = 1 - 0.7375 = 0.2625$$

$$\text{Now } \sqrt{\frac{PQ}{n}} = \sqrt{\frac{0.7375 \times 0.2625}{80}} = 0.049$$

Confidence interval is $\left(P - z_{\alpha/2} \sqrt{\frac{PQ}{n}}, P + z_{\alpha/2} \sqrt{\frac{PQ}{n}} \right)$ where $z_{\alpha/2} = 2.58$

$$\text{or } \left(P - 3 \sqrt{\frac{PQ}{n}}, P + 3 \sqrt{\frac{PQ}{n}} \right)$$

i.e., $(0.7375 - 3 \times 0.049, 0.7375 + 3 \times 0.049)$ i.e., $(0.59, 0.88)$

Example 16 : In a random sample of 100 packages shipped by air freight 13 had some damage. Construct 95% confidence interval for the true proportion of damage package.

[JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Here p = Sample proportion of damage packages = $\frac{13}{100} = 0.13$

$$\therefore q = 1 - p = 1 - 0.13 = 0.87$$

$$\text{S.E. of } p = \sqrt{\frac{PQ}{n}} = \sqrt{\frac{pq}{n}} \quad (\because P \text{ is not known, we take } p \text{ for } P)$$

$$= \sqrt{\frac{0.13 \times 0.87}{100}} = 0.034$$

$\therefore$ 95% confidence limits for the population proportion of P of damage packages are $p \pm 1.96 (\text{S.E. of } p) = 0.13 \pm 1.96 (0.034) = 0.13 \pm 0.067$

i.e., the confidence limits are 0.063 and 0.197

Hence the 95% confidence interval for the true proportion of damage packages is $(0.063, 0.197)$.

Example 17 : In a hospital 480 females and 520 male babies were born in a week. Do these figures confirm the hypothesis that males and females are born in equal number?

[JNTU (K) May 2010 (Set No. 1)]

Solution : Null Hypothesis H_0 : The probability of equal proportion i.e. $P = 0.5$

Alternative Hypothesis H_1 : The proportion is not equal i.e., $P \neq 0.5$

$$n = \text{Total number of births} = 480 + 520 = 1000$$

$$p = \text{Proportion of females born} = \frac{480}{1000} = 0.48$$

$$\text{Population proportion} = 0.5 = P, Q = 1 - P = 1 - 0.5 = 0.5$$

The test statistic is

$$z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.48 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = \frac{-0.02}{0.0158} = -1.265$$

$\therefore |z| = 1.265$. We use two-tailed test.

Since $|z| = 1.265 < 1.96$, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the males and females are born in equal proportions.

Example 18 : A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipments received 80 were faulty. Test the claim at 0.05 level.

[JNTU (H) Nov. 2010 (Set No. 3)]

Solution : We have $P = 95\% = 0.95, n = 200$

$$\therefore Q = 1 - P = 1 - 0.95 = 0.05$$

$$p = \text{Observed proportion of faulty pieces} = \frac{80}{200} = 0.4$$

Let the Null Hypothesis be $H_0 : P = 0.95$

The Alternative Hypothesis is $H_1 : P < 0.95$

Since H_1 is one-tailed (left-tailed), we use one-tailed (left-tailed) test.

Let us assume that H_0 is true.

$$\begin{aligned} \text{The test statistic is } z &= \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.4 - 0.95}{\sqrt{\frac{0.95 \times 0.05}{200}}} \\ &= \frac{-0.55}{0.0154} = -0.0085 \end{aligned}$$

Thus we see that $z = -0.0085 > -1.645$

Hence, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the claim of the manufacturer is justified.

Example 19 : A random sample of 500 Apples was taken from a large consignment of 60 were found to be bad, obtain the 98% confidence limits for the percentage number of bad apples in the consignment.

[JNTU (H) Dec. 2011 (Set No. 2), (A) Dec. 2017]

Solution : We are given

$n = \text{Sample size} = 500$

$x = \text{Number of bad apples} = 60$

$$\therefore p = \text{Proportion of bad apples} = \frac{x}{n} = \frac{60}{500} = 0.12$$

Hence $q = 1 - p = 1 - 0.12 = 0.88$

$$\therefore \text{Standard error of } p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.12 \times 0.88}{500}} = 0.014$$

The 98% confidence limits for p are

$$\left(p - z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}, p + z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} \right) \text{ where } z_{\alpha/2} = 2.33$$

$$= (0.12 - 2.33 \times 0.014, 0.12 + 2.33 \times 0.014)$$

$$= (0.087, 0.153)$$

Thus the confidence limits for percentage of bad apples

$$= (0.087 \times 100, 0.153 \times 100) = (8.7, 15.3)$$

Example 20 : In a study designed to investigate whether certain detonators used with explosives in coal mining meet the requirement that at least 90% will ignite the explosive when charged. It is found that 174 of 200 detonators function properly. Test the null hypothesis $P = 0.9$ against the alternative hypothesis $P \neq 0.9$ at the 0.05 level of significance.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let the Null Hypothesis be $H_0 : P = 0.9$

Then the Alternative Hypothesis is $H_1 : P \neq 0.9$

Since H_1 is two - tailed, we use two - tailed test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

If p is the observed proportion of detonators functioning properly, then $p = \frac{174}{200} = 0.87$

$Q = 1 - P = 1 - 0.9 = 0.1$ and $n = 200$

$$\therefore z = \frac{0.87 - 0.9}{\sqrt{\frac{0.9 \times 0.1}{200}}} = \frac{-0.03}{\sqrt{\frac{0.09}{200}}} = -1.41$$

$$\therefore |z| = 1.41 < 1.645$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant evidence to say that the given kind of detonators fails to meet the required standard.

9.19 TEST FOR EQUALITY OF TWO PROPORTIONS (OR TEST OF SIGNIFICANCE OF DIFFERENCE BETWEEN TWO SAMPLE PROPORTIONS - LARGE SAMPLES)

Let p_1 and p_2 be the sample proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2 .

To test whether the two samples have been drawn from the same population.

1. The Null Hypothesis $H_0 : P_1 = P_2$
2. The Alternative Hypothesis $H_1 : P_1 \neq P_2$
3. The Test Statistic : There are two ways of computing a test statistic z .

(a) When the population proportions P_1 and P_2 are known.

In this case, $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$ and p_1, p_2 are sample proportions.

$$\therefore \text{Standard Error of Difference} = \text{S. E. } (p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

(b) When the population proportions P_1 and P_2 are not known but sample proportions p_1 and p_2 are known.

In this case we have two methods to estimate P_1 and P_2 .

(i) **Method of Substitution** : In this method, sample proportions p_1 and p_2 are substituted for P_1 and P_2 .

$$\therefore \text{S. E. } (p_1 - p_2) = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

(ii) **Method of Pooling** : In this method, the estimated value for the two population proportions is obtained by pooling the two sample proportions p_1 and p_2 into a single proportion p by the formula given below.

Sample proportion of two samples or estimated value of p is given by

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \text{ so that } q = 1 - p$$

$$\therefore \text{S. E. } (p_1 - p_2) = \sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}} = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Rejection Rule for $H_0 : P_1 = P_2$.

- (i) If $|z| > 1.96$, then reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, then reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, then reject H_0 at 10% level of significance.

SOLVED EXAMPLES

Example 1 : Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level. [JNTU (H) May 2011 (Set No. 2)]

Solution : Given sample sizes, $n_1 = 400$, $n_2 = 600$

Proportion of men, $p_1 = \frac{200}{400} = 0.5$

Proportion of women, $p_2 = \frac{325}{600} = 0.541$

1. **Null hypothesis H_0 :** Assume that there is no significant difference between the option of men and women as far as proposal of flyover is concerned.

i.e., $H_0 : p_1 = p_2 = p$

2. **Alternative hypothesis $H_1 : p_1 \neq p_2$ (two tailed)**

3. **The test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ [Method of pooling]**

where $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

$$= \frac{400 \times \frac{200}{400} + 600 \times \frac{325}{600}}{400 + 600} = \frac{525}{1000} = 0.525$$

and $q = 1 - p = 1 - 0.525 = 0.475$

$$\therefore z = \frac{0.5 - 0.541}{\sqrt{0.525 \times 0.475 \left(\frac{1}{400} + \frac{1}{600} \right)}} = \frac{-0.041}{0.032} = -1.28$$

$$|z| = 1.28$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance.

i.e., there is no difference of opinion between men and women as far as proposal of flyover is concerned.

Example 2 : Random samples of 400 men and 200 women in a locality were asked whether they would like to have a bus stop near their residence. 200 men and 40 women in favour of the proposal. Test the significance between the difference of two proportions at 5% level.

[JNTU (H) Dec. 2011 (Set No. 1)]

Solution : Let P_1 and P_2 be the population proportions in a locality who favour the bus stop.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Here n_1 = Number of men in I sample = 400

n_2 = Number of women in II sample = 200

x_1 = Number of men in favour of the proposal = 200

x_2 = Number of women in favour of the proposal = 40

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{200}{400} = \frac{1}{2} \text{ and } p_2 = \frac{x_2}{n_2} = \frac{40}{200} = \frac{1}{5}$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 40}{400 + 200} = \frac{240}{600} = \frac{2}{5}$$

$$\therefore q = 1 - p = 1 - \frac{2}{5} = \frac{3}{5}$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{1}{2} - \frac{1}{5}}{\sqrt{\frac{2}{5} \times \frac{3}{5} \left(\frac{1}{400} + \frac{1}{200} \right)}} = \frac{0.3}{0.0424} = 7.07 > 1.96$$

Since $z > 1.96$, we reject the Null Hypothesis at 5% level of significance and conclude that the difference between the two proportions is highly significant i.e., there is difference between the men and women in their attitude towards the bus stop near their residence.

Example 3 : A manufacturer of electronic equipment subjects samples of two completing brands of transistors to an accelerated performance test. If 45 of 180 transistors of the first kind and 34 of 120 transistors of the second kind fail the test, what can he conclude at the level of significance $\alpha = 0.05$ about the difference between the corresponding sample proportions ?

[JNTU 2003S, 2004 (Set No. 1)]

Solution : We have $n_1 = 180, x_1 = 45, x_2 = 34, n_2 = 120$

$$\text{and } p_1 = \frac{x_1}{n_1} = \frac{45}{180} = 0.25, p_2 = \frac{x_2}{n_2} = \frac{34}{120} = 0.283$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{45 + 34}{180 + 120} = \frac{79}{300} = 0.263$$

$$q = 1 - p = 1 - 0.263 = 0.737$$

1. Null Hypothesis $H_0 : p_1 = p_2$ i.e., there is no difference
2. Alternative Hypothesis $H_1 : p_1 \neq p_2$ i.e., there is a difference
3. Level of significance : $\alpha = 0.05$
4. The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad [\text{Method of pooling}]$$

$$= \frac{0.25 - 0.283}{\sqrt{(0.263)(0.737) \left(\frac{1}{180} + \frac{1}{120} \right)}}$$

$$= \frac{-0.033}{\sqrt{(0.194)(0.0134)}} = \frac{-0.033}{0.051} = -0.647$$

$$\therefore |z| = 0.647$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance i.e., the difference between the proportions is not significant.

Example 4 : On the basis of their total scores, 200 candidates of a civil service examination are divided into two groups, the upper 30% and the remaining 70%. Consider the first question of the examination. Among the first group, 40 had the correct answer, whereas among the second group, 80 had the correct answer. On the basis of these results, can one conclude that the first question is not good at discriminating ability of the type being examined here ?

[JNTU 2003S, 2004, JNTU (K) Nov. 2009 (Set No. 2)]

Solution : We have $n_1 = 60, n_2 = 140, x_1 = 40, x_2 = 80$

$$\text{and } p_1 = \frac{x_1}{n_1} = \frac{40}{60} = \frac{2}{3} = 0.667, p_2 = \frac{x_2}{n_2} = \frac{80}{140} = 0.571$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{40 + 80}{60 + 140} = \frac{120}{200} = 0.6$$

$$\text{and } q = 1 - p = 1 - 0.6 = 0.4$$

1. Null Hypothesis $H_0 : p_1 = p_2$
2. Alternative Hypothesis $H_1 : p_1 \neq p_2$
3. Level of significance : $\alpha = 0.05$ (assumed)
4. The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad [\text{Method of pooling}]$$

$$= \frac{0.667 - 0.571}{\sqrt{(0.6)(0.4) \left(\frac{1}{60} + \frac{1}{40} \right)}} = \frac{0.096}{0.0756} = 1.27$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance i.e., the first question is good enough in discriminating the ability of the students of both groups.

Note : Suppose the population proportions P_1 and P_2 are given and $P_1 \neq P_2$. If we want to test the hypothesis that the difference $(P_1 - P_2)$ in population proportions is likely to be hidden in simple samples of sizes n_1 and n_2 from the two populations respectively, then

$$z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{\left(\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2} \right)}}$$

If the sample proportions are not known, then we use

$$|z| = \frac{|P_1 - P_2|}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

Example 5 : A cigarette manufacturing firm claims that its brand A line of cigarettes outsells its brand B by 8%. If it is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another sample of 100 smokers prefer brand B, test whether the 8% difference is a valid claim.

Solution : Here $n_1 = 200$, $n_2 = 100$,

$$p_1 = \frac{42}{200} = 0.21, \quad p_2 = \frac{18}{100} = 0.18$$

$$\text{and } P_1 - P_2 = 8\% = \frac{8}{100} = 0.08$$

1. **Null Hypothesis H_0** : Assume that 8% difference in the sale of two brands of cigarettes is a valid claim *i.e.*,

$$H_0 : P_1 - P_2 = 0.08$$

2. **Alternative Hypothesis H_1** : $P_1 - P_2 \neq 0.08$ (Two tail test)

3. The test statistic is, $z = \frac{(p_1 - p_2) - (P_1 - P_2)}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ [Method of pooling]

$$\text{Now } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{200(0.21) + 100(0.18)}{200 + 100} = \frac{42 + 18}{300} = 0.2$$

$$\text{and } q = 1 - 0.2 = 0.8$$

$$\therefore z = \frac{0.03 - 0.08}{\sqrt{0.2 \times 0.8 \left(\frac{1}{200} + \frac{1}{100} \right)}} = \frac{-0.05}{0.0489} = -1.02$$

$$\therefore |z| = 1.02$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance. *i.e.*, 8% difference in the sale of two brands of cigarettes is a valid claim.

Example 6 : In two large populations, there are 30%, and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations. [JNTU (A) Nov. 2010, (K) Nov. 2009, May 2013 (Set No. I)]

Solution :

$$\text{Given } n_1 = 1200, n_2 = 900$$

$$P_1 = \text{Proportion of fair haired people in the first population} = \frac{30}{100} = 0.3$$

$$P_2 = \text{Proportion of fair haired people in the second population} = \frac{25}{100} = 0.25$$

1. **Null Hypothesis H_0** : Assume that the sample proportions are equal *i.e.*, the difference in population proportions is likely to be hidden in sampling *i.e.*,

$$H_0 : P_1 = P_2$$

2. **Alternative Hypothesis H_1** : $P_1 \neq P_2$

3. The test statistic is, $z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$

$$\text{where } Q_1 = 1 - P_1 = 1 - 0.3 = 0.7$$

$$Q_2 = 1 - P_2 = 1 - 0.25 = 0.75$$

$$\therefore z = \frac{0.3 - 0.25}{\sqrt{\frac{0.3 \times 0.7}{1200} + \frac{0.25 \times 0.75}{900}}} = \frac{0.05}{0.0195} = 2.56$$

$$\text{i.e., } z = 2.56$$

Since $z > 1.96$, therefore we reject the Null Hypothesis H_0 at 5% level of significance (Two-tailed test) i.e., the sample proportions are not equal. Thus we conclude that the difference in population proportions is unlikely that the real difference will be hidden.

Example 7 : A company wanted to introduce a new plan of work and a survey was conducted for this purpose. Out of sample of 500 workers in one group 62% favoured the new plan and another group of sample of 400 workers 41% were against the new plan. Is there any significant difference between the two groups in their attitude towards the new plan at 5% level of significance?

[JNTU (A) Dec. 2009 (Set No. 1)]

Solution : Given $n_1 = 500, n_2 = 400, p_1 = 0.62$ and $p_2 = 1 - 0.41 = 0.59$.

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{500(0.62) + 400(0.59)}{500 + 400} = \frac{310 + 236}{900} = \frac{546}{900} = 0.607$$

$$\text{and } q = 1 - p = 1 - 0.607 = 0.393.$$

1. Null Hypothesis $H_0 : p_1 = p_2$ i.e., there is no significant difference between the two groups in their attitude towards the new plan.

2. Alternative Hypothesis $H_1 : p_1 \neq p_2$ (two tailed test)

3. Level of significance : $\alpha = 0.05$

$$\begin{aligned} \text{4. The test statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.62 - 0.59}{\sqrt{(0.607)(0.393) \left(\frac{1}{500} + \frac{1}{400} \right)}} \\ &= \frac{0.03}{\sqrt{0.2385 \times 0.45}} = \frac{0.03}{0.3276} = 0.0916 \end{aligned}$$

Since $|z| < 1.96$, the null hypothesis is accepted at 5% level of significance. Hence, we conclude that there is no significant difference between the two groups in their attitude towards the new plan.

Example 8 : In a city A, 20% of a random sample of 900 school boys has a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys has the same defect. Is the difference between the proportions significant at 0.05 level of significance.

[JNTU (K) Nov. 2009 (Set No. 1)]

Solution : We are given

$$n_1 = \text{First sample size} = 900$$

$$n_2 = \text{Second sample size} = 1600$$

$$x_1 = 20\% \text{ of } 900 = 180$$

$$x_2 = 18.5\% \text{ of } 1600 = 296$$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{180}{900} = 0.2, \quad p_2 = \frac{x_2}{n_2} = \frac{296}{1600} = 0.185$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 296}{900 + 1600} = \frac{476}{2500} = 0.19$$

$$\text{Now } q = 1 - p = 1 - 0.19 = 0.81$$

1. Null Hypothesis $H_0 : p_1 = p_2$

2. Alternative Hypotheses $H_1 : p_1 \neq p_2$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\begin{aligned} \text{i.e., } z &= \frac{0.2 - 0.185}{\sqrt{(0.19)(0.81) \left(\frac{1}{900} + \frac{1}{1600} \right)}} = \frac{-0.015}{\sqrt{(0.1539)(0.001736)}} \\ &= \frac{-0.015}{0.01634} = -0.918 \end{aligned}$$

$$\therefore |z| = 0.918$$

Since $|z| < 1.96$, we accept the null hypothesis H_0 at 5% level of significance i.e., there is no significant difference between the proportions.

Example 9 : In a random sample of 1000 persons from town A, 400 are found to be consumers of wheat. In a sample of 800 from town B, 400 are found to be consumers of wheat. Do these data reveal a significant difference between town A and town B, so far as the proportion of wheat consumers is concerned? [JNTU (H) Nov. 2009, (K) May 2012 (Set No. 2)]

Solution : We have

n_1 = sample size of town A = 1000

n_2 = sample size of town B = 800

x_1 = Number of consumers of wheat from town A = 400

x_2 = Number of consumers of wheat from town B = 400

p_1 = Proportion of consumers of wheat in town A = $\frac{x_1}{n_1} = \frac{400}{1000} = 0.4$

p_2 = Proportion of consumers of wheat in town B = $\frac{x_2}{n_2} = \frac{400}{800} = 0.5$

$$\therefore p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400 + 400}{1000 + 800} = \frac{800}{1800} = \frac{8}{18} = \frac{4}{9} \text{ and } q = 1 - p = 1 - \frac{4}{9} = \frac{5}{9}.$$

1. Null Hypothesis $H_0 : p_1 = p_2$ i.e., there is no difference.

2. Alternative Hypothesis $H_1 : p_1 \neq p_2$ i.e., there is a difference.

3. Level of significance : $\alpha = 0.05$

$$\begin{aligned} 4. \text{ The test statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.4 - 0.5}{\sqrt{\frac{4}{9} \times \frac{5}{9} \left(\frac{1}{1000} + \frac{1}{800} \right)}} \\ &= \frac{-0.1}{\sqrt{\frac{20}{81} \times \frac{1800}{(1000)(800)}}} = -4.242 \end{aligned}$$

$$\therefore |z| = 4.2424$$

$$\text{Also } z_\alpha = 1.96$$

Since $|z| > z_\alpha$, we reject the null hypothesis H_0 at 5% level of significance.

$\therefore$ There is significant difference between town A and town B, as the proportion of wheat consumers is concerned.

Example 10 : In a sample of 600 students of a certain college 400 are found to use ball pens. In another college from a sample of 900 students 450 were found to use ball pens. Test whether 2 colleges are significantly different with respect to the habit of using ball pens.

[JNTU (H) Nov. 2010 (Set No. 1)]

Solution : Let P_1 and P_2 be the population proportions of students who use ball pens in two colleges.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Given x_1 = Number of students who use ball pens in first college = 400

n_1 = Sample size of first sample = 600

x_2 = Number of students who use ball pens in second college = 450

n_2 = Sample size of second sample = 900

$$\therefore p_1 = \text{Proportion of students who use ball pens in first college} = \frac{x_1}{n_1} = \frac{400}{600} = \frac{2}{3}$$

$$p_2 = \text{Proportion of students who use ball pens in second college} = \frac{x_2}{n_2} = \frac{450}{900} = \frac{1}{2}$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{400 + 450}{600 + 900} = \frac{850}{1500} = \frac{17}{30} = 0.57$$

$$q = 1 - p = 1 - \frac{17}{30} = \frac{13}{30} = 0.43$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{2}{3} - \frac{1}{2}}{\sqrt{(0.57)(0.43) \left(\frac{1}{600} + \frac{1}{900} \right)}} = \frac{1}{6 \sqrt{0.2451(0.0027)}} = \frac{1}{0.1565} = 6.39$$

$$\therefore z = 6.39 > 1.96$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference between the two colleges as far as using ball pens habit is concerned.

Example 11 : 100 articles from a factory are examined and 10 are found to be defective. 500 similar articles from a second factory are found to be 15 defective. Test the significance between the difference of two proportions at 5% level. [JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let P_1 and P_2 be the proportions of defective articles in the population of articles manufactured by the two factories.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

$$\text{Here } n_1 = 100, n_2 = 500, p_1 = \frac{10}{100} = 0.1 \text{ and } p_2 = \frac{15}{500} = 0.03$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{100(0.1) + 500(0.03)}{100 + 500} = \frac{10 + 15}{600} = \frac{25}{600} = \frac{1}{24} = 0.042$$

$$q = 1 - p = 1 - 0.042 = 0.958$$

$$\text{and S.E. of } (p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.042)(0.958) \left(\frac{1}{100} + \frac{1}{500} \right)} = 0.022$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.1 - 0.03}{0.022} = \frac{0.07}{0.022} = 3.18$$

$$\therefore z = 3.18 > 1.96$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference between the two proportions.

Example 12 : A machine produced 20 defective articles in a batch of 400. After overhauling it produced 10 defectives in a batch of 300. Has the machine being improved after overhauling? [JNTU(K) Nov.2011 (Set No.4)]

Solution : If p_1 and p_2 be the sample proportions of imperfect items put out by the machine before and after the overhauling and P_1 and P_2 be the corresponding population proportions.

Here $n_1 = 400$, $x_1 = 20$, $n_2 = 300$ and $x_2 = 10$.

$$\text{Now } p_1 = \frac{20}{400} = \frac{1}{20}, p_2 = \frac{10}{300} = \frac{1}{30}$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{30}{700} = \frac{3}{70}$$

$$\therefore q = 1 - p = 1 - \frac{3}{70} = \frac{67}{70}$$

1. **Null Hypothesis:** $H_0 : p_1 = p_2$

2. **Alternative Hypothesis:** $H_1 : p_1 > p_2$. It leads to a one-tailed test.

3. **Level of significance:** $\alpha = 0.05$

$$\begin{aligned} \text{4. Test Statistic is } z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{\frac{1}{20} - \frac{1}{30}}{\sqrt{\frac{3}{70} \times \frac{67}{70} \left(\frac{1}{400} + \frac{1}{300} \right)}} \\ &= \frac{0.0167}{\sqrt{0.041(0.0058)}} = \frac{0.0167}{0.0154} = 1.0844 \end{aligned}$$

5. **Critical value:** The critical value of z at $\alpha = 0.05$ is $z_{0.05} = 1.645$ for one tailed test

6. **Decision.** Since $|z| (= 1.0844) < z_{0.05} (= 1.645)$, therefore, the Null Hypothesis is accepted

i.e., There is no significant difference.

Hence, the machine has not improved after overhauling.

Example 13 : A machine puts out 9 imperfect articles in a sample of 200 articles. After the Machine is overhauled it puts out 5 imperfect articles in a sample of 700 articles. Test at 5% level whether the Machine is improved? [JNTU (H) Nov. 2010 (Set No. 3)]

Solution : Let P_1 and P_2 be the proportions of imperfect articles in the population of articles manufactured by the machine before and after overhauling, respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

We have $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ and S.E. of $(p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

Here $n_1 = 200, p_1 = \frac{9}{200} = 0.045, n_2 = 700, p_2 = \frac{5}{700} = 0.00714$

$$\therefore p = \frac{0.045 \times 200 + 0.00714 \times 700}{200 + 700} = \frac{9 + 5}{900} = \frac{14}{900} = 0.0156$$

and $q = 1 - p = 1 - 0.0156 = 0.9844$

$$\therefore \text{S.E. of } (p_1 - p_2) = \sqrt{0.0156 \times 0.9844 \left(\frac{1}{200} + \frac{1}{700} \right)} = 0.0099$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.045 - 0.00714}{0.0099} = \frac{0.03786}{0.0099} = 3.824$$

Since $z = 3.824 > 3$, we reject the Null Hypothesis at 5% level of significance and conclude that the machine is improved.

Example 14 : A machine puts out 16 imperfect articles in a sample of 500 articles. After the Machine is overhauled it puts out 3 imperfect articles in a sample of 100 articles. Has the Machine improved? [JNTU (H) Dec. 2011 (Set No. 3)]

Solution : Let P_1 and P_2 be the proportions of imperfect articles in the population of articles manufactured by the machine before and after overhauling, respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 > P_2$

Here,

n_1 = Sample size before the machine overhauling = 500

x_1 = No. of imperfect articles before overhauling = 16

n_2 = Sample size after the machine overhauling = 100

x_2 = No. of imperfect articles after overhauling = 3

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{16}{500} = 0.032 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{3}{100} = 0.03 \text{ and } p_1 > p_2$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 3}{500 + 100} = \frac{19}{600} = 0.032$$

and $q = 1 - p = 0.968$

Since H_1 is one-tailed (right-tailed), we use one-tailed test (right-tailed test).

Let us assume that H_0 is true. The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.032 - 0.03}{\sqrt{0.032 \times 0.968\left(\frac{1}{500} + \frac{1}{100}\right)}}$$

$$= \frac{0.002}{0.019} = 0.104$$

Thus we see that $z = 0.104 < 1.645$. Hence we accept the Null Hypothesis H_0 at 5% level of significance and conclude that the machine has improved.

Example 15 : In an investigation on the machine performance the following results are obtained.

	No. of units inspected	No. of defectives
Machine 1	375	17
Machine 2	450	22

Test whether there is any significant performance of two machines at $\alpha = 0.05$

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let P_1 and P_2 be the proportions of defective units in the population of units inspected in Machine 1 and Machine 2 respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

$$p_1 = \frac{17}{375} = 0.045, \quad p_2 = \frac{22}{450} = 0.049 \quad \text{and} \quad p_1 > p_2$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} \quad \text{and S.E. of } (p_1 - p_2) = \sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\text{Here } n_1 = 375, n_2 = 450, p_1 = 0.045, p_2 = 0.049$$

$$\therefore p = \frac{375 \times 0.045 + 450 \times 0.049}{375 + 450} = \frac{17 + 22}{375 + 450} = \frac{39}{825} = 0.047$$

$$q = 1 - p = 1 - 0.047 = 0.953$$

$$\therefore \text{S.E. of } (p_1 - p_2) = \sqrt{0.047 \times 0.953\left(\frac{1}{375} + \frac{1}{450}\right)} = 0.015$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.045 - 0.049}{0.015} = -0.267$$

$$\therefore |z| = 0.267 < 1.96$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference in performance of two machines.

Example 16 : Among the items produced by a factory out of 500, 15 were defective, in another sample out of 400, 20 were defective. Test the significance between the differences of two proportions at 5% level. [JNTU (H) Dec. 2011 (Set No. 2)]

Solution : Let P_1 and P_2 be the proportions of defective items in the population of two sample items produced by the factory.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$.

Here $n_1 = 500, n_2 = 400, x_1 = 15$ and $x_2 = 20$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{15}{500} = \frac{3}{100} = 0.03 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{20}{400} = \frac{1}{20} = 0.05$$

$$\text{We have } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{15 + 20}{500 + 400} = \frac{35}{900} = 0.039$$

$$\text{Now } q = 1 - p = 1 - 0.039 = 0.961$$

Assuming that H_0 is true, the test statistic is

$$\begin{aligned} z &= \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.03 - 0.05}{\sqrt{0.039 \times 0.961 \left(\frac{1}{500} + \frac{1}{400} \right)}} \\ &= \frac{-0.02}{0.013} = -1.54 \end{aligned}$$

$$\therefore |z| = 1.54 < 1.96$$

Hence we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two proportions.

Example 17 : Among the items produced by a factory out of 800, 65 were defective in another sample out of 300, 40 were defective. Test the significance between the differences of two proportions at 1% level. [JNTU (H) Apr. 2012 (Set No. 2), (A) Dec. 2017]

Solution : Let P_1 and P_2 be the proportions of defective items in the population of items produced by a factory in two samples respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 < P_2$

Since H_1 is one - tailed (left - tailed), we use one - tailed test (left - tailed test)

Here $n_1 = 800, x_1 = 65, p_1 = \frac{65}{800} = 0.081, n_2 = 300, x_2 = 40, p_2 = \frac{40}{300} = 0.1333$

and $p_1 < p_2$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{65 + 40}{800 + 300} = \frac{105}{1100} = 0.095$$

and $q = 1 - p = 1 - 0.095 = 0.905$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.081 - 0.1333}{\sqrt{0.095 \times 0.905 \left(\frac{1}{800} + \frac{1}{300} \right)}} \\ = \frac{-0.0523}{\sqrt{0.086 \times 0.00459}} = \frac{-0.0523}{0.01986} = -2.6324$$

Thus we see that $z = -2.6324 < -2.33$

Hence, we reject the Null Hypothesis H_0 at 1% level of significance and conclude that there is a significant difference between the two proportions.

Example 18 : Before an increase on excise duty on tea 500 people out of a sample of 900 found to have the habit of having tea. After an increase on excise duty 250 are have the habit of having tea among 1100. Is there any decrease in the consumption of tea. Test at 5% level. [JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let P_1 and P_2 be the population proportions of people who have the habit of tea before and after an increase on excise duty on tea respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

The Alternative Hypothesis is $H_1 : P_1 > P_2$

Now $p_1 = \frac{500}{900} = 0.556, p_2 = \frac{250}{1100} = 0.227$

We have $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ and S.E. of $(p_1 - p_2) = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

Here $n_1 = 900, n_2 = 1100, p_1 = 0.556, p_2 = 0.227$

$$\therefore p = \frac{900 \times 0.556 + 1100 \times 0.227}{900 + 1100} = 0.375$$

$$q = 1 - p = 1 - 0.375 = 0.625$$

$$\therefore \text{S.E. of } (p_1 - p_2) = \sqrt{0.375 \times 0.625 \left(\frac{1}{900} + \frac{1}{1100} \right)} = 0.022$$

Assuming that H_0 is true, the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E. of } (p_1 - p_2)} = \frac{0.556 - 0.227}{0.022} = 14.95$$

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference in the consumption of tea before and after an increase on excise duty on tea.

Example 19 : During a country wide investigation the incidence of tuberculosis was found to be 1%. In a college of 400 students 3 reported to be affected, where as in another college of 1200 students 10 were affected. Does this indicate any significant difference?

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : Let P_1 and P_2 be the population proportions of students who are affected tuberculosis in two colleges respectively.

Let the Null Hypothesis be $H_0 : P_1 = P_2$

Then the Alternative Hypothesis is $H_1 : P_1 \neq P_2$

Here $n_1 = 400, x_1 = 3, n_2 = 1200, x_2 = 10$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{3}{400} = 0.0075 \text{ and } p_2 = \frac{x_2}{n_2} = \frac{10}{1200} = 0.0083$$

Given $p = 0.01$ and $q = 1 - p = 0.99$

Assuming that H_0 is true, the test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\therefore z = \frac{0.0075 - 0.0083}{\sqrt{0.01 \times 0.99 \left(\frac{1}{400} + \frac{1}{1200} \right)}} = \frac{-0.0008}{0.0057} = -0.14$$

$$\therefore |z| = 0.14 < 1.96$$

Hence, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two colleges as far as the incidence of tuberculosis concerned.

Example 20 : A random sample of 300 shoppers at a supermarket includes 204 who regularly use cents off coupons. In another sample of 500 shoppers at a supermarket includes 75 who regularly use cents off coupons

(a) Construct confidence interval for the probability that any one shopper at the supermarket, selected at random, will regularly use cents off coupons.

- (b) Test the significance between the difference of two proportions at 2% level.

[JNTU (H) Apr. 2012 (Set No. 4)]

Solution : Here we are given

$$n_1 = 300, x_1 = 204, n_2 = 500 \text{ and } x_2 = 75$$

$$p_1 = \text{Proportion of shoppers who use cents off coupons in the first sample} = \frac{204}{300} = 0.68$$

$$p_2 = \text{Proportion of shoppers who use cents off coupons in the second sample} \\ = \frac{75}{500} = 0.15$$

- (a) The 98% confidence interval for the probability that any one shopper in sample selected at random is

$$\left(p_1 - z_{\alpha/2} \cdot \sqrt{\frac{p_1 q_1}{n_1}}, p_1 + z_{\alpha/2} \cdot \sqrt{\frac{p_1 q_1}{n_1}} \right) \text{ where } q_1 = 1 - p_1 = 1 - 0.68 = 0.32$$

$$\text{i.e., } \left(0.68 - (2.33) \sqrt{\frac{0.68 \times 0.32}{300}}, 0.68 + (2.33) \sqrt{\frac{0.68 \times 0.32}{300}} \right)$$

$$\text{i.e., } (0.68 - 0.063, 0.68 + 0.063) \text{ or } (0.62, 0.74)$$

- (b) Here we set up the Null Hypothesis H_0 that $p_1 = p_2$

i.e., the sample proportions are equal

i.e., there is no difference in the two proportions.

Then the Alternative Hypothesis is $H_1 : p_1 \neq p_2$

$$\text{Now } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{204 + 75}{300 + 500} = \frac{279}{800} = 0.35$$

$$\therefore q = 1 - p = 1 - 0.35 = 0.65$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \\ = \frac{0.68 - 0.15}{\sqrt{0.35 \times 0.65 \left(\frac{1}{300} + \frac{1}{500} \right)}} \\ = \frac{0.53}{0.035} = 15.14$$

Since $z = 15.14 > 2.58$, we reject the Null Hypothesis at 1% level of significance, and conclude that there is a significant difference in the two proportions.

Example 21 : A sample poll of 300 voters from district A and 200 voters from district B showed that 56% and 48% respectively, were in favour of a given candidate. At a 0.05 level of significance, test the hypothesis that there is a difference in the districts.

[JNTU (A) April 2012 (Set No. 3)]

Solution : We are given $n_1 = 300, n_2 = 200$ and $P_1 = \frac{56}{100} = 0.56, P_2 = \frac{48}{100} = 0.48$

1. Null Hypothesis $H_0 : P_1 = P_2$ i.e., There is no significant difference in the districts.
2. Alternative Hypothesis $H_1 : P_1 \neq P_2$
3. The level of significance, $\alpha = 0.05$
4. The test statistic is

$$z = \frac{P_1 - P_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}} \quad \text{where } Q_1 = 1 - P_1 = 1 - 0.56 = 0.44$$

$$\text{and } Q_2 = 1 - P_2 = 1 - 0.48 = 0.52$$

$$\therefore z = \frac{0.56 - 0.48}{\sqrt{\frac{0.56 \times 0.44}{300} + \frac{0.48 \times 0.52}{200}}} = \frac{0.08}{0.045} = 1.78$$

Since $z < 1.96$, we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference between the two districts towards the favour of a given candidate.

Example 22 : A study shows that 16 of 200 tractors produced on one assembly line required extensive adjustments before they could be shipped, while the same was true for 14 of 400 tractors produced on another assembly line. At the 0.01 level of significance, does this support the claim that the second production line does superior work?

[JNTU (A) April 2012 (Set No. 3)]

Solution : We are given

n_1 = sample size of first sample = 200

n_2 = sample size of second sample = 400

x_1 = number of tractors on first assembly line which required adjustments = 16.

x_2 = number of tractors on second assembly line which required adjustments = 14.

$$\text{Now } p_1 = \frac{x_1}{n_1} = \frac{16}{200} = 0.08 \quad \text{and} \quad p_2 = \frac{x_2}{n_2} = \frac{14}{400} = 0.035$$

$$\therefore p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 14}{200 + 400} = \frac{30}{600} = \frac{1}{20} = 0.05$$

$$q = 1 - p = 1 - 0.05 = 0.95$$

Null Hypothesis $H_0 : p_1 = p_2$

Alternative Hypothesis $H_1 : p_1 < p_2$

Level of significance $\alpha = 0.05$

The test statistic is

$$z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.08 - 0.035}{\sqrt{0.05 \times 0.95 \left(\frac{1}{200} + \frac{1}{400} \right)}} \\ = \frac{0.045}{\sqrt{0.0475 \times 0.0075}} = \frac{0.045}{0.0189} = 2.384$$

Since $z > 1.96$, we reject the Null Hypothesis at 5% level and conclude that the second production is not superior than first.

Example 23 : If 120 out of 200 patients suffering from a certain disease are cured by allopathy and 240 out of 500 patients are cured by homeopathy, is there reason enough to believe that allopathy is better than homeopathy in curing the disease? Use $\alpha = 0.05$ LOS.

[JNTU (A) Dec. 2014]

Solution : Let p_1, p_2 respectively be the proportions of patients suffering from a certain disease are cured by allopathy and homeopathy.

We are given $n_1 = 200, x_1 = 120, n_2 = 500$ and $x_2 = 240$

$$\text{Now } p_1 = \frac{x_1}{n_1} = \frac{120}{200} = 0.6, p_2 = \frac{x_2}{n_2} = \frac{240}{500} = 0.48$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 240}{200 + 500} = \frac{360}{700} = 0.51$$

$$\therefore q = 1 - p = 1 - 0.51 = 0.49$$

1. Null Hypothesis is $H_0 : p_1 = p_2$

i.e. there is no difference between allopathy and homeopathy

2. Alternative Hypothesis $H_1 : p_1 > p_2 \Rightarrow$ It leads to a one-tailed test

i.e. Allopathy is better than homeopathy

3. Level of significance, $\alpha = 0.05$

4. Test statistic is $z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

$$\text{i.e. } z = \frac{0.60 - 0.48}{\sqrt{0.51 \times 0.49 \left(\frac{1}{200} + \frac{1}{500}\right)}} = 2.9$$

5. **Critical value** : The critical value of z at $\alpha = 0.05$ is $z_{0.05} = 1.645$ for one-tailed test

6. **Decision** : Since the calculated value of $|z| (= 2.9) > z_{0.05} (= 1.645) \Rightarrow$ that the null hypothesis is rejected

$\Rightarrow$ Allopathy is better than homeopathy in curing particular disease.

REVIEW QUESTIONS

- Write the Test Statistic :
 - in the test of significance for single mean.
 - for the test of significance for single proportion and
 - for the test of significance for difference of proportion.

EXERCISE

- In a random sample of 400 persons from a large population, 120 are females. Can it be said that males and females are in the ratio 5 : 3 in the population ? use 1% level of significance.
- A coin is tossed 900 times and heads appear 490 times. Does this result support the hypothesis that the coin is unbiased.
 - A coin is tossed 10,000 times and it turns up head 5195 times. Test the hypothesis that the coin is unbiased. Use a 0.01 level of significance.
 - A die is tossed 9000 times and it falls with 5 or 6 upwards 3240 times. Test the hypothesis that the die is unbiased. Use a 0.01 level of significance.
- A wholesaler in apples claims that only 4% of the apples supplied by him are defective. A random sample of 600 apples contained 36 defective apples. Test the claim of the wholesaler.
- In a sample of 500 people in Tamil Nadu 280 are tea drinkers and the rest are coffee drinkers. Can we assume that both coffee and tea are equally popular in this state at 1% level of significance.

5. A manufacturer claimed that at least 98% of the steel pipes which he supplied to a factory conformed to specifications. An examination of a sample of 500 pieces of pipes revealed that 30 were defective. Test his claim at a significance level 5%.
6. The machine puts out 16 imperfect articles in a sample of 500. After machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved?
7. In a sample of 600 students of a certain college 400 are found to use ball pens. In another college, from a sample of 900 students 450 were found to use ball pens. Test whether the two colleges are significantly different with respect to the habit of using ball pens. [JNTU (H) Nov. 2010 (Set No. 1)]
8. In a certain district A, 450 persons were considered regular consumers of tea out of a sample of 1000 persons. In another district B, 400 were regular consumers of tea out of a sample of 800 persons. Do these facts reveal a significant difference between the two districts as far as tea drinking habit is concerned?
9. (i) A machine produced 10 defective articles in a batch of 200. After overhauling it produced 4 defectives in a batch of 100. Has the machine been improved.
(ii) A machine puts out 21 defective articles in a sample of 500 articles. Another machine gives 3 defective articles in a sample of 100. Are the two machines significantly different in their performance?
(iii) A machine puts out 10 imperfect articles in a sample of 200. After the machine is overhauled it puts out 4 imperfect articles in a batch of 100. Has the machine been improved? [JNTU (K) May 2012 (Set No. 1)]
10. A random sample of 100 articles selected from a batch of 2,000 articles shows that the average diameter of the articles = 0.354 with a S.D. = 0.048. Find 95% confidence interval for the average of this batch of 2,000 articles.
11. A sample of 100 iron bars is said to be drawn from a large number of bars whose lengths are normally distributed with mean 4 feet and S.D. 0.6 feet. If the sample mean is 4.2 feet, can the sample be regarded as a truly random sample.
12. The mean life of a sample of 100 electric bulbs produced by a company is found to be 1570 hrs with a S.D. of 120 hrs. If μ is the mean life time of all the bulbs produced by the company, test the hypothesis $\mu = 1600$ hrs against the alternative hypothesis $\mu \neq 1600$ hrs at 5% level of significance.
13. A sample of 400 male students is found to have a mean height of 171.38 cm. Can it be reasonably regarded as a sample from a large population with mean height of 171.17 cm and S.D. 3.30 cm.
14. A sample of 100 workers in a large plant gave a mean assembly time of 294 seconds with a S.D. of 12 seconds in a time and motion study. Find a 95% confidence interval for the mean assembly time for all the workers in the plant.

15. The mean breaking strength of the cables supplied by a manufacturer is 1800 with a *S.D.* 100. By a new technique in the manufacturing process, it is claimed that the breaking strength of the cables have increased. In order to test this claim, a sample of 50 cables is tested. It is found that the mean breaking strength is 1850. Can we support the claim at 1% level of significance.
16. An investigation of the relative merits of two kinds of flash light batteries showed that a random sample of 100 batteries of brand A tested on the average 36.5 hrs with a *S.D.* of 1.8 hrs while a random sample of 80 batteries of brand B tested on the average 36.8 hrs with a *S.D.* of 1.5 hrs. Use a level of significance of 0.05 to test whether the observed difference between the average life times is significant.
17. Given the following information relating to two places *A* and *B*. Test whether there is any significant difference between their mean wages :

	<i>A</i>	<i>B</i>
Mean wages (Rs.)	47	49
<i>S.D.</i> (Rs.)	28	40
Number of workers	1000	1500

18. The mean consumption of food grains among 400 sampled middle class consumers is 380 gms per day per person with a *S.D.* of 120 gms. A similar sample survey of 600 working class consumers gave a mean of 410 gms with a *S.D.* of 80 gms. Are we justified in saying that the two classes consume the same quantity of food grains. Use 5% level of significance.
19. In a city 250 men out of 750 were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers.

[JNTU 2005S, 2006S, 2008S (Set No. 1)]

20. Test the significance of the difference between the means of the sample from the following data :

	Size of sample	Mean	<i>S.D.</i>
Sample A	100	61	4
Sample B	200	63	6

21. In a random sample of 100 tube lights produced by company A, the mean life time of tube light is 1190 hours with standard deviation of 90 hours. Also in a random sample of 75 tube lights from company B the mean life time is 1230 hours with standard deviation of 120 hours. Is there a difference between the mean life times of the two brands of tube lights at a significance level of 0.05?
22. 500 articles from a factory are examined and found to be 2% defective. 800 similar articles from a second factory are found to have only 1.5 % defective. Can it reasonably be concluded that the products of the first factory are inferior to those of second ?

[JNTU (K) Nov. 2011 (Set No. 2)]

ANSWERS

1. $Z = -3.125$; The males and females in the population are not in the ratio 5 : 3.
2. (i) $H_0 : P = \frac{1}{2}$ (coin is unbiased), $Z = 2.39$, rejected at 5% level.
 (ii) $H_0 : P = \frac{1}{2}$ (coin is unbiased), $Z = 3.9$, rejected at 1% level.
 (iii) $H_0 : P = \frac{1}{3}$ (die is unbiased), $Z = 5.4$, rejected at 1% level.
3. $H_0 : P = 0.04$; $H_1 : P > 0.04$, $Z = 10$, highly significant.
4. $H_0 : P = \frac{1}{2}$, $H_1 : P \neq \frac{1}{2}$, $Z = 2.68$, significant at 1% level.
5. $H_0 : P = 0.98$, $H_1 : P < 0.98$ (left-tail test), $Z = 6.38$, H_0 rejected.
6. $H_0 : P_1 = P_2$, $H_1 : P_2 < P_1$, $Z = 1.04$, not significant.
7. $H_0 : P_1 = P_2$; $H_1 : P_1 \neq P_2$; $|Z| = 6.48$, significant.
8. $H_0 : P_1 = P_2$, $H_1 : P_1 \neq P_2$, $|Z| = 2.08$, significant at 5% level.
9. (i) $H_0 : P_1 = P_2$, $H_1 : P_1 > P_2$, $|Z| = 0.3846$, not significant.
 (i.e., not improved the performance of the machine)
10. 0.34, 0.36.
11. $H_0 : \mu = 4$ ft, $H_1 : \mu \neq 4$ ft, $Z = 3.33$ significant.
12. $|Z| = 2.5$, significant at 5% level of significance.
14. 291.64, 296.35.
15. $H_0 : \mu = 1800$, $\sigma = 100$, $H_1 : \mu_1 > 1800$, $Z = 3.53$, Accept H_0
16. $H_0 : \bar{x}_1 = \bar{x}_2$; $|Z| = 1.21$, Accept H_0
17. $H_0 : \bar{x}_1 = \bar{x}_2$, $|Z| < 1.96$, Accept H_0
18. $H_0 : \bar{x}_1 = \bar{x}_2$, $H_1 : \bar{x}_1 \neq \bar{x}_2$, $|Z| = 4.39$ significant.
19. $H_0 : p = \frac{1}{3}$, $H_1 : P < 0.5$; $z = -9.128$, H_0 rejected.
21. $|z| = 2.42$